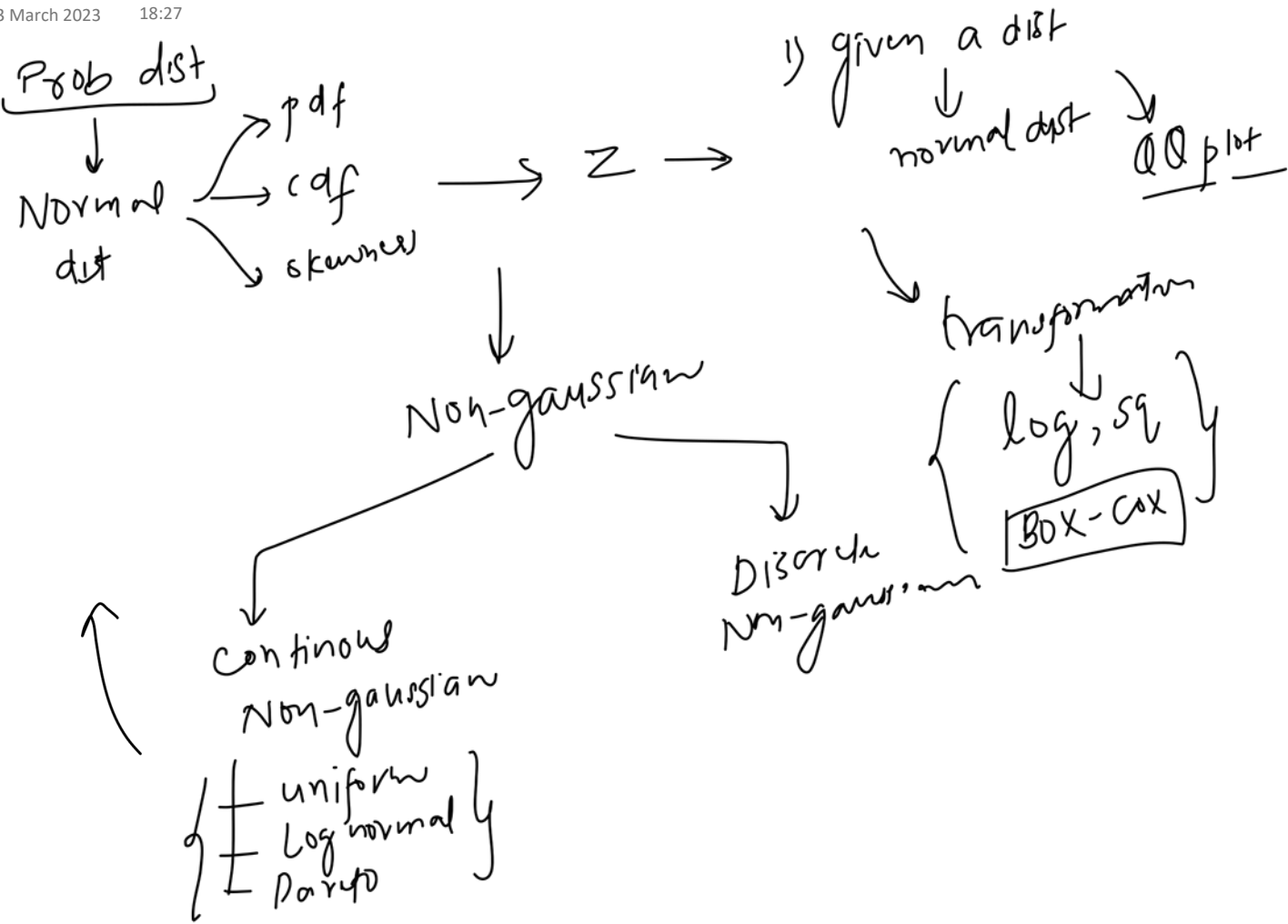


Non Gaussian Distributions

Recap

23 March 2023 18:27

https://youtu.be/U6QCc_3zgUk?si=QC1SAV9IPuDIDGRk

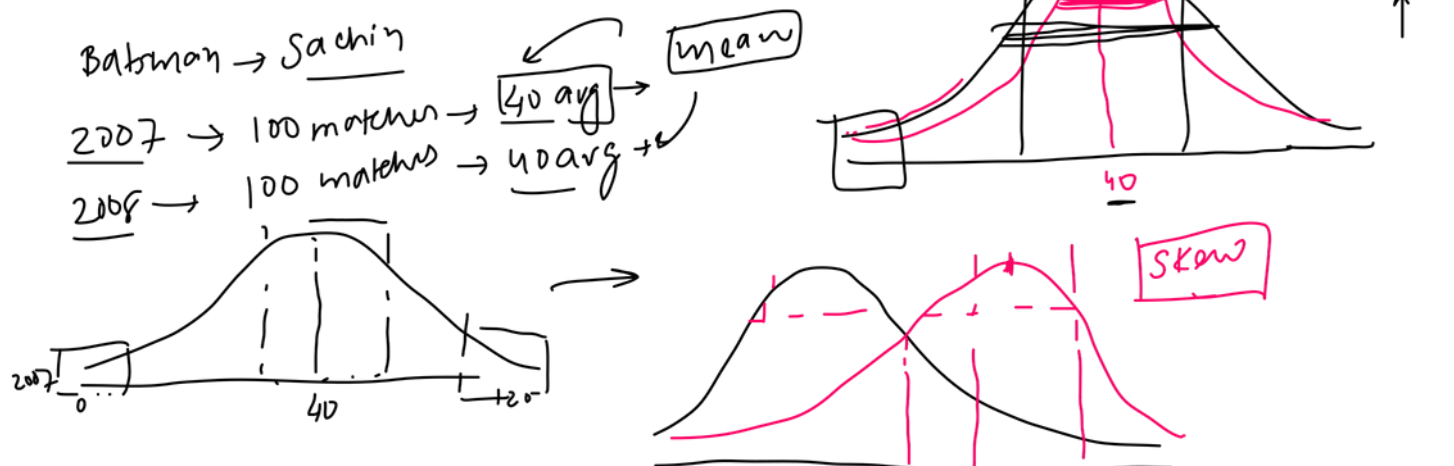


Kurtosis

23 March 2023 13:18

•What is Kurtosis? Tailedness (fatness of tails)

Kurtosis is the 4th statistical moment. In probability theory and statistics, kurtosis (meaning "curved, arching") is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Like skewness, kurtosis describes a particular aspect of a probability distribution.



It tells the density of outliers with the help of fatness of tails.

•False notation about Kurtosis

<https://en.wikipedia.org/wiki/Kurtosis>

•Formula

sample kurtosis

$$\left\{ \frac{n * (n+1)}{(n-1) * (n-2) * (n-3)} * \sum_i^n \left(\frac{x_i - \bar{x}}{s} \right)^4 \right\} - \frac{3 * (n-1)^2}{(n-2) * (n-3)}$$

•Practical Use-case

In finance, kurtosis risk refers to the risk associated with the possibility of extreme outcomes or "fat tails" in the distribution of returns of a particular asset or portfolio.

If a distribution has high kurtosis, it means that there is a higher likelihood of extreme events occurring, either positive or negative, compared to a normal distribution.

In finance, kurtosis risk is important to consider because it indicates that there is a greater probability of large losses or gains occurring, which can have significant implications for investors. As a result, investors may want to adjust their investment strategies to account for kurtosis risk.

•Excess Kurtosis & Types

Excess kurtosis is a measure of how much more peaked or flat a distribution is



Excess Kurtosis & Types

Excess kurtosis is a measure of how much more peaked or flat a distribution is compared to a normal distribution, which is considered to have a kurtosis of 0. It is calculated by subtracting 3 from the sample kurtosis coefficient.

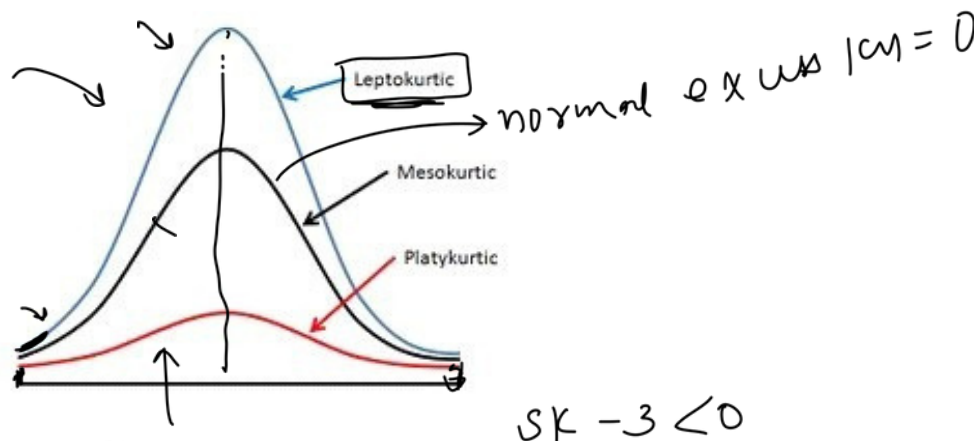
Types of Kurtosis

$$s_{amp} k - 3 > 0$$

Leptokurtic

A distribution with positive excess kurtosis is called leptokurtic. "Lepto-" means "slender". In terms of shape, a leptokurtic distribution has fatter tails. This indicates that there are more extreme values or outliers in the distribution.

Example - Assets with positive excess kurtosis are riskier and more volatile than those with a normal distribution, and they may experience sudden price movements that can result in significant gains or losses.



Platykurtic

A distribution with negative excess kurtosis is called platykurtic. "Platy-" means "broad". In terms of shape, a platykurtic distribution has thinner tails. This indicates that there are fewer extreme values or outliers in the distribution.

Assets with negative excess kurtosis are less risky and less volatile than those with a normal distribution, and they may experience more gradual price movements that are less likely to result in large gains or losses.

Mesokurtic

Distributions with zero excess kurtosis are called mesokurtic. The most prominent example of a mesokurtic distribution is the normal distribution family, regardless of the values of its parameters.

$$(\mu, \sigma)$$

Mesokurtic is a term used to describe a distribution with a excess kurtosis of 0, indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.

Example -

In finance, a mesokurtic distribution is considered to be the ideal distribution for assets or portfolios, as it represents a balance between risk and return.

•How to find if a given distribution is normal or not?

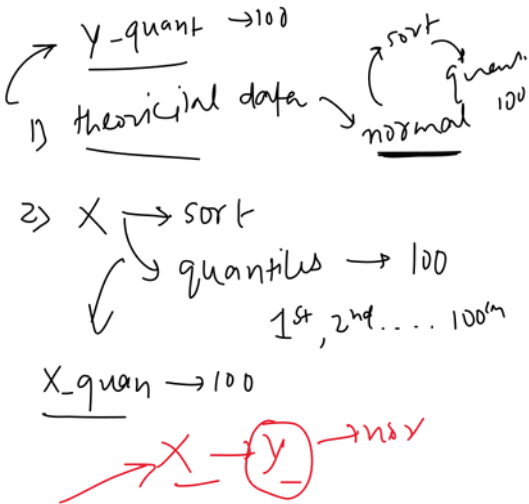
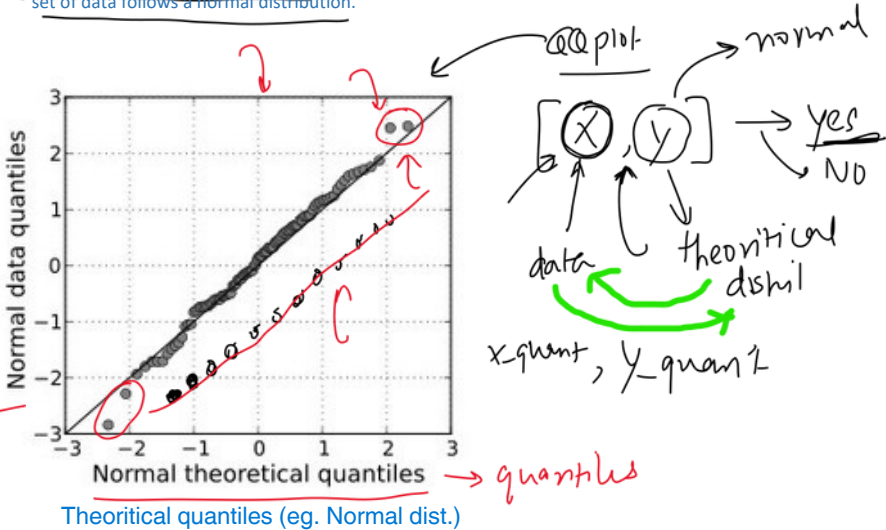
- Visual inspection: One of the easiest ways to check for normality is to visually inspect a histogram or a density plot of the data. A normal distribution has a bell-shaped curve which means that the majority of the data falls in the middle, and the tails taper off symmetrically. If the distribution looks approximately bell-shaped, it is likely to be normal. QQ Plot: Another way to check for normality is to create a normal probability plot (also known as a Q-Q plot) of the data. A normal probability plot plots the observed data against the expected values of a normal distribution. If the data points fall along a straight line, the distribution is likely to be normal.



Statistical tests: There are several statistical tests that can be used to test for normality, such as the Shapiro-Wilk test, the Anderson-Darling test, and the Kolmogorov-Smirnov test. These tests compare the observed data to the expected values of a normal distribution and provide a p-value that indicates whether the data is likely to be normal or not. A p-value less than the significance level (usually 0.05) suggests that the data is not normal.

•What is a QQ Plot and how is it plotted?

A QQ plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of two sets of data. It is particularly useful for determining whether a set of data follows a normal distribution.

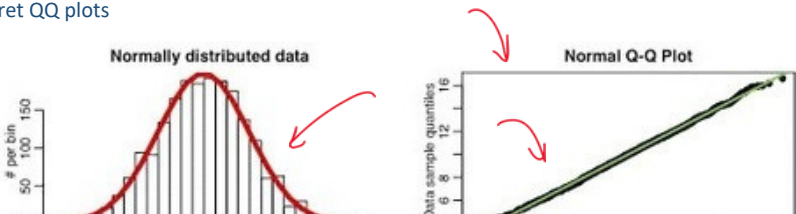


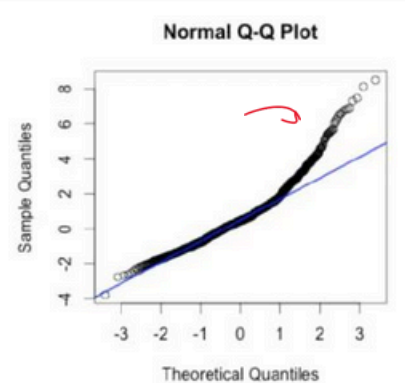
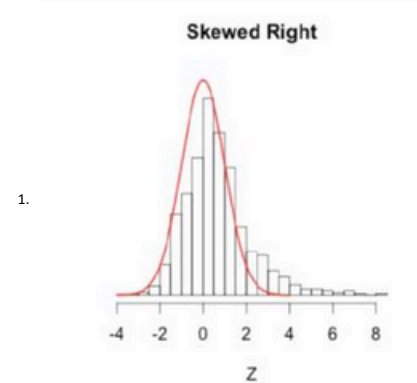
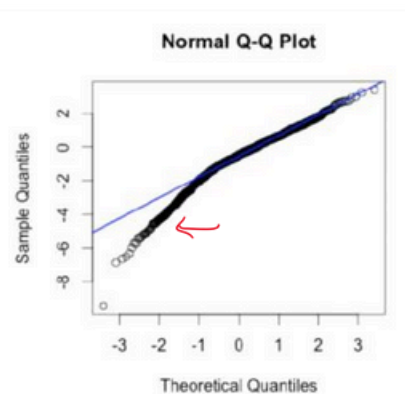
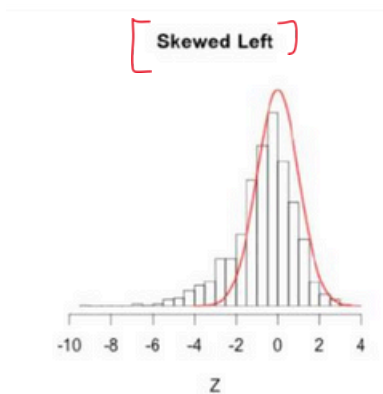
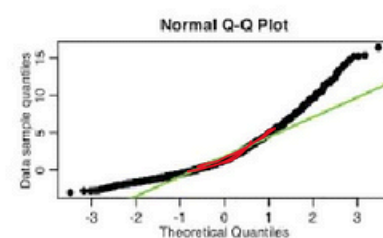
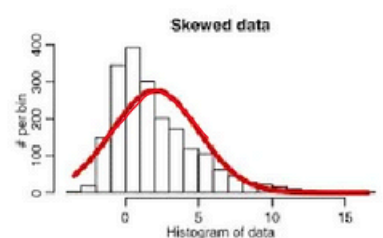
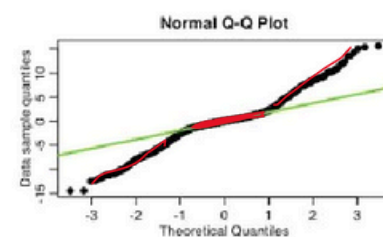
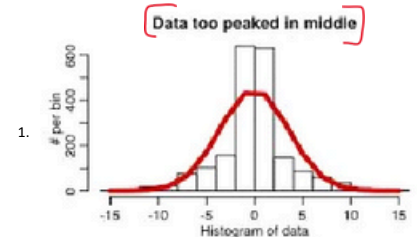
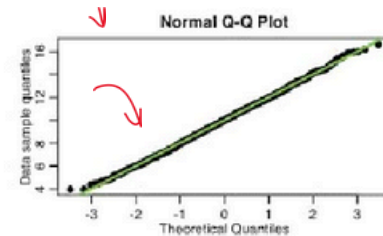
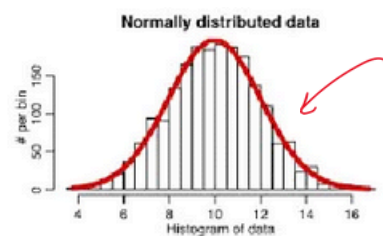
In a QQ plot, the quantiles of the two sets of data are plotted against each other. The quantiles of one set of data are plotted on the x-axis, while the quantiles of the other set of data are plotted on the y-axis. If the two sets of data have the same distribution, the points on the QQ plot will fall on a straight line. If the two sets of data do not have the same distribution, the points will deviate from the straight line.

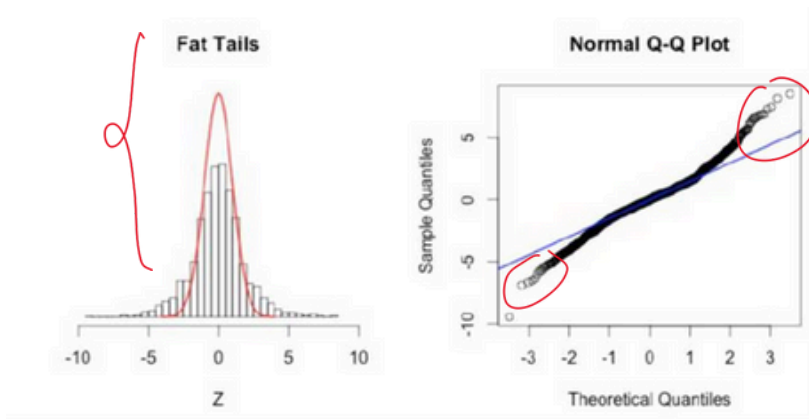
•Python example

<https://www.statsmodels.org/dev/generated/statsmodels.graphics.gofplots.qqplot.html>

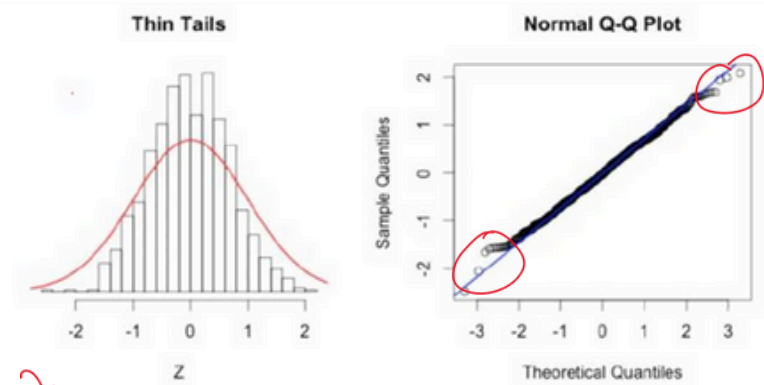
•How to interpret QQ plots



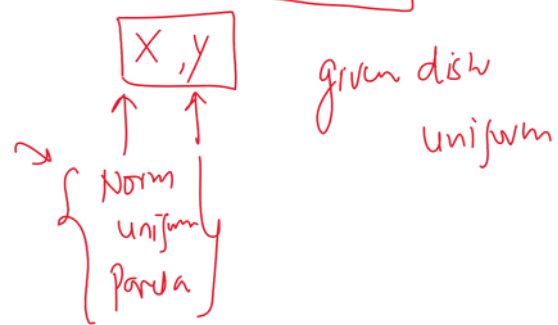




1.



• Does QQ plot only detect normal distribution?



Uniform Distribution

23 March 2023 13:19

• What is Uniform Distribution and its types

In probability theory and statistics, a uniform distribution is a probability distribution where all outcomes are equally likely within a given range. This means that if you were to select a random value from this range, any value would be as likely as any other value.

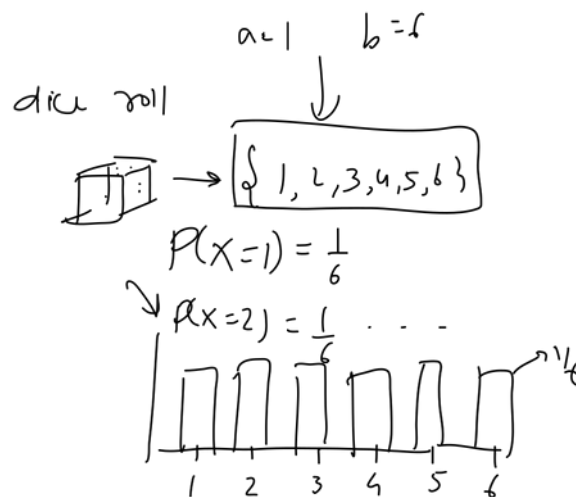
Types



Denoted as

$$X \sim U(a, b) \rightarrow \text{parameters}$$

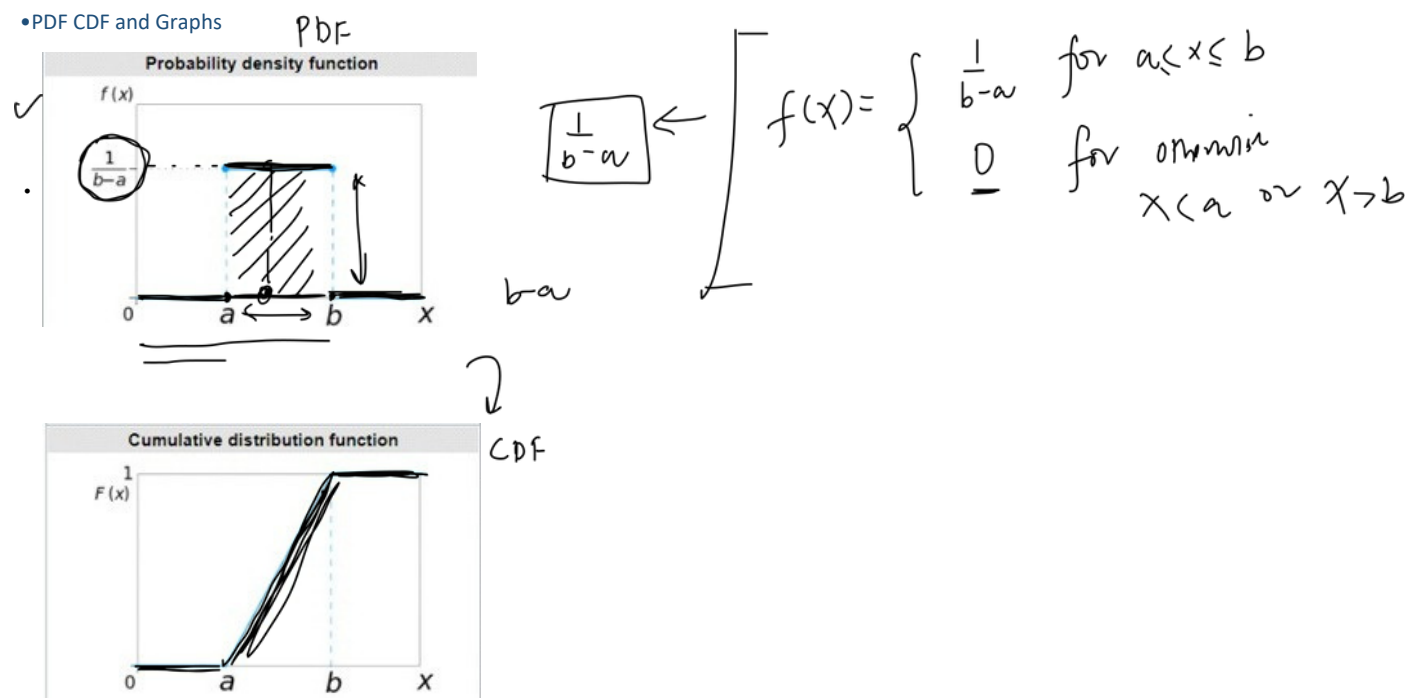
$a \leftrightarrow b$
 lower high



• Examples

- The height of a person randomly selected from a group of individuals whose heights range from 5'6" to 6'0" would follow a continuous uniform distribution.
- The time it takes for a machine to produce a product, where the production time ranges from 5 to 10 minutes, would follow a continuous uniform distribution.
- The distance that a randomly selected car travels on a tank of gas, where the distance ranges from 300 to 400 miles, would follow a continuous uniform distribution.
- The weight of a randomly selected apple from a basket of apples that weighs between 100 and 200 grams, would follow a continuous uniform distribution.

• PDF CDF and Graphs

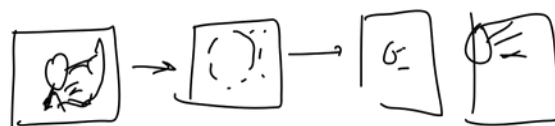


https://en.wikipedia.org/wiki/Continuous_uniform_distribution

- Skewness $\rightarrow 0 \rightarrow \text{symmetric} \rightarrow \text{Normal}$

• Application in Machine learning and Data Science

a. **Random initialization:** In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of

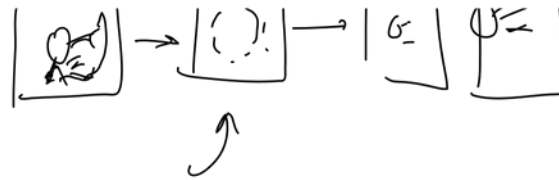


a. **Random initialization:** In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.

b. **Sampling:** Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.

c. **Data augmentation:** In some cases, you may want to artificially increase the size of your dataset by generating new examples that are similar to the original data. Uniform distribution can be used to generate new data points that are within a specified range of the original data.

d. **Hyperparameter tuning:** Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.

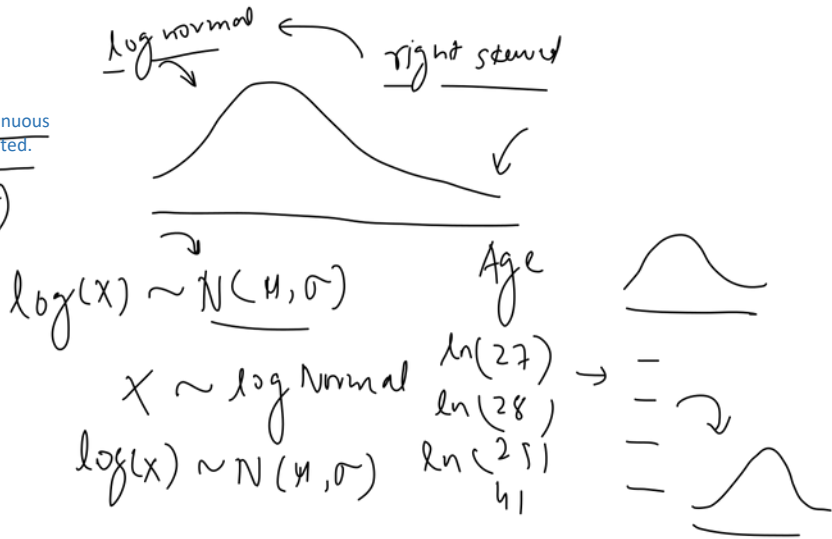
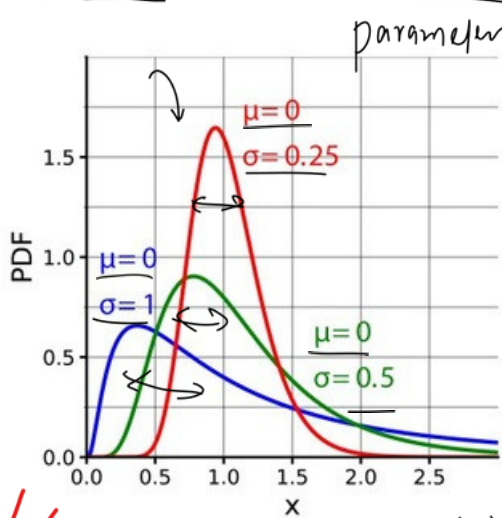


→ deep learning

Log Normal Distribution

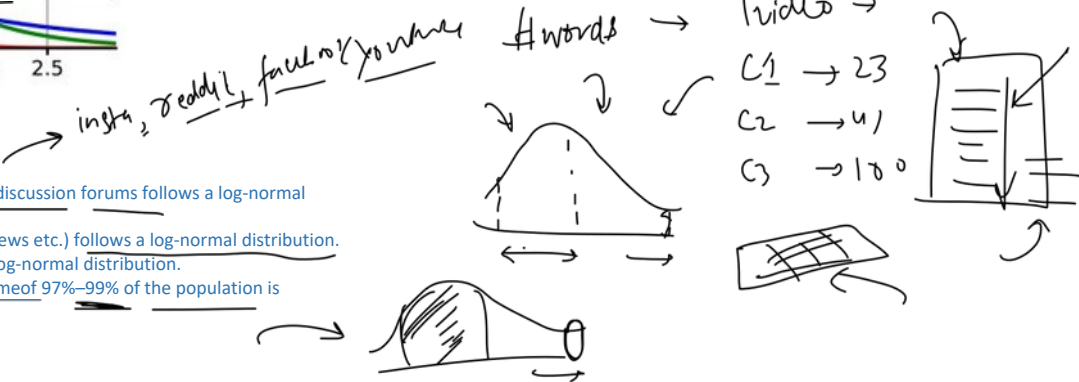
23 March 2023 13:19

In probability theory and statistics, a lognormal distribution is a heavy tailed continuous probability distribution of a random variable whose logarithm is normally distributed.



Examples

- The length of comments posted in Internet discussion forums follows a log-normal distribution.
- Users' dwell time on online articles (jokes, news etc.) follows a log-normal distribution.
- The length of chess games tends to follow a log-normal distribution.
- In economics, there is evidence that the income of 97%-99% of the population is distributed log-normally.



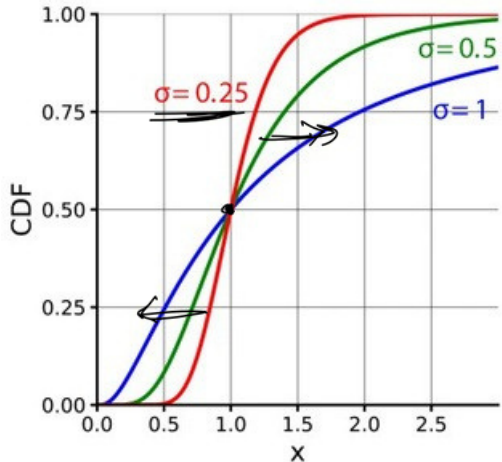
Denoted as

$$X \sim \text{lognormal}(\mu, \sigma) \rightarrow \ln(x) \sim N(\mu, \sigma)$$

PDF Equation

$$\frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} \leftarrow \text{similar Normal dist.}$$

CDF



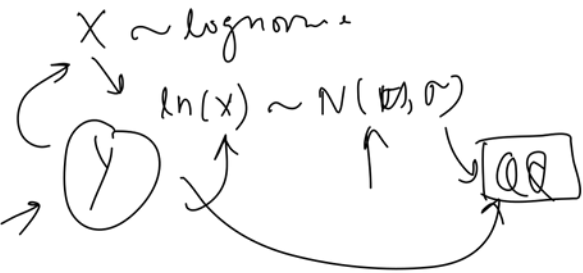
Skewness

skewed

How to check if a random variable is log normally distributed?

Skewness skewness

How to check if a random variable is log normally distributed?



Pareto Distribution

23 March 2023 13:19

Pareto Distribution

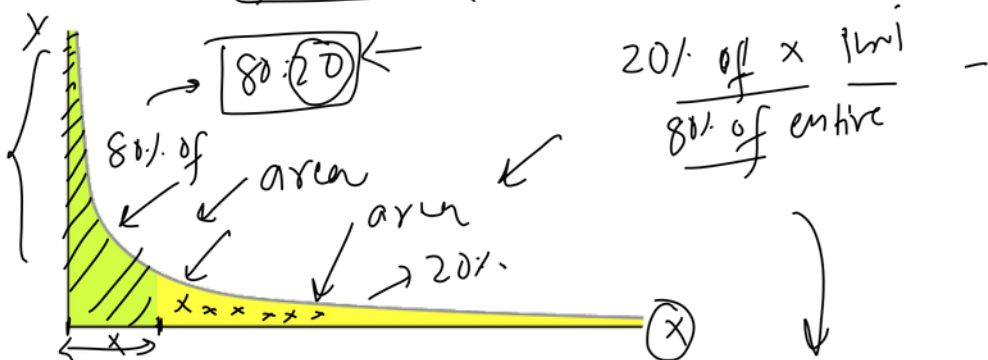
The Pareto distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income, and other quantities that exhibit a similar power-law behaviour

What is Power Law

In mathematics, a power law is a functional relationship between two variables, where one variable is proportional to a power of the other. Specifically, if y and x are two variables related by a power law, then the relationship can be written as:

$y = k \cdot x^a$

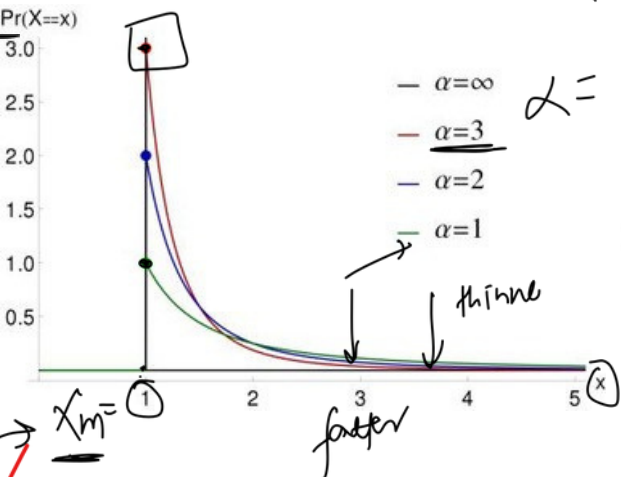
$y = k x^a$



Vilfredo Pareto originally used this distribution to describe the allocation of wealth among individuals since it seemed to show rather well the way that a larger portion of the wealth of any society is owned by a smaller percentage of the people in that society. He also used it to describe distribution of income. This idea is sometimes expressed more simply as the Pareto principle or the "80-20 rule" which says that 20% of the population controls 80% of the wealth

↑ application

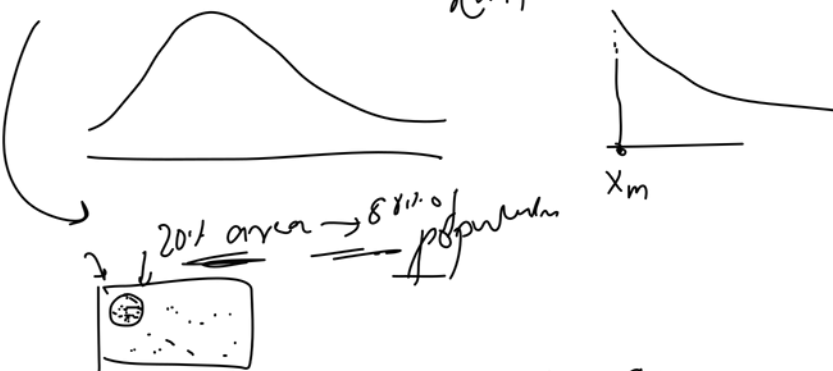
Graph & Parameters



$X \sim P(\alpha)$

$\alpha = 1.1$

pdf = $\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$



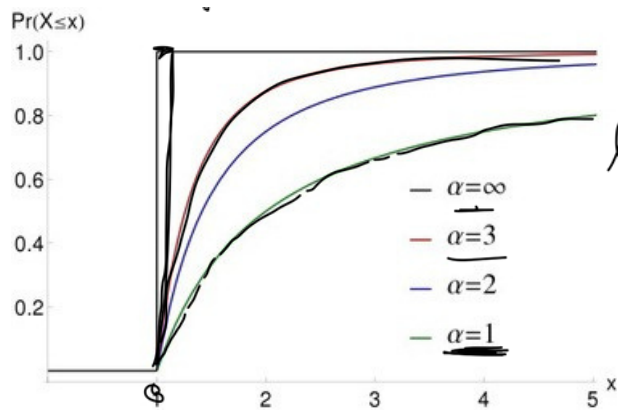
Examples

- The sizes of human settlements (few cities, many hamlets/villages)
- File size distribution of Internet traffic which uses the TCP protocol (many smaller files, few larger ones)

80% of entire go
↓
20% of file
80%
20% of 80%

CDF

$Pr(X \leq x)$

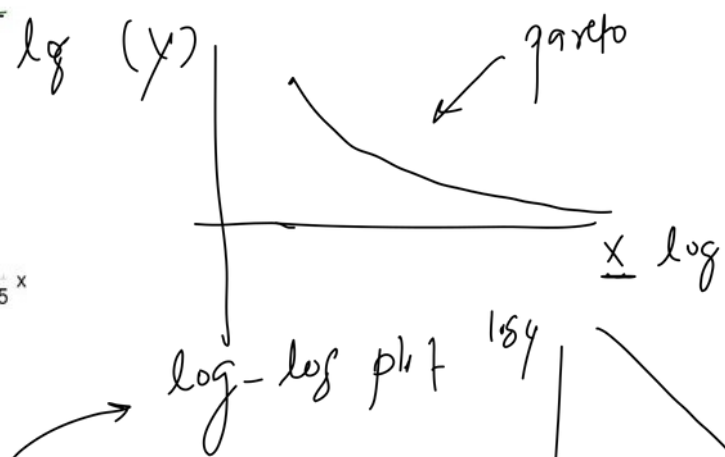


Skewness → skewed

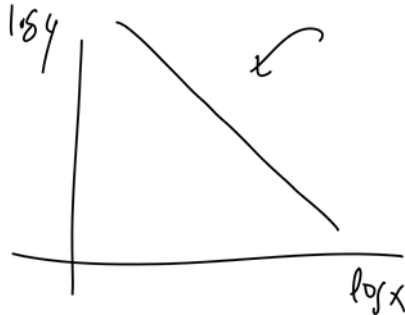
[How to detect if a distribution is Pareto Distribution?]

QQ plot

X, Y →
↑
pareto

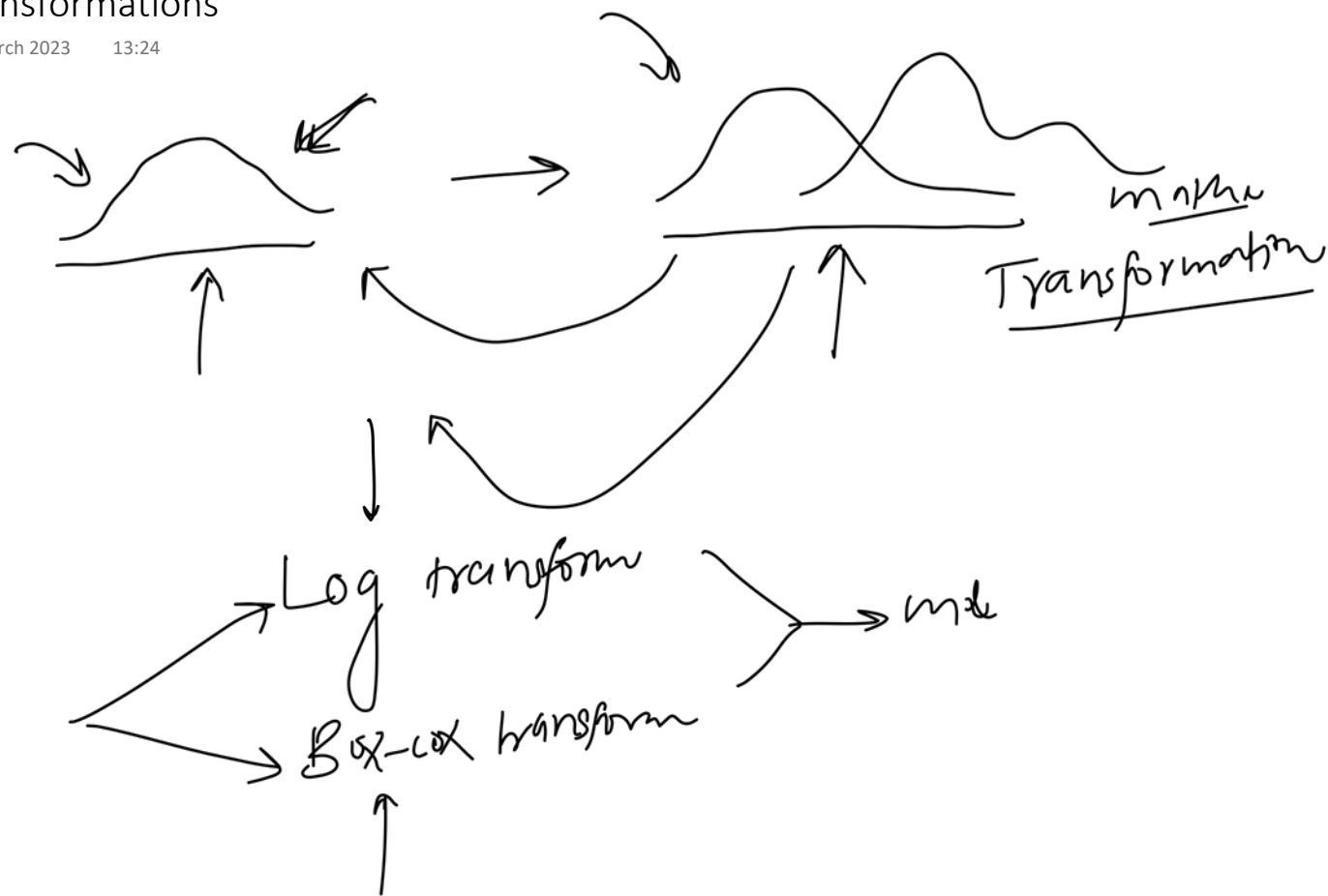


$$y = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$



Transformations

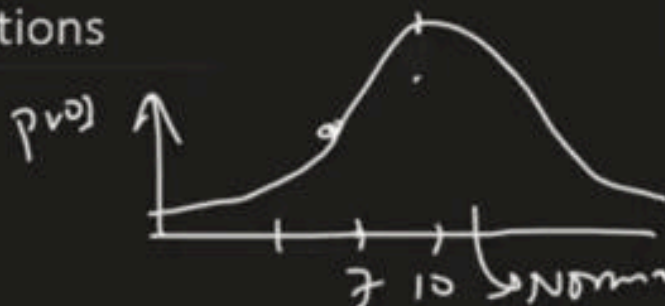
23 March 2023 13:24



Mathematical Transformations

Friday, April 16, 2021

5:21 PM



FE

feature transformations



Mathematical trans

Normal distribution

- 1) log trans
- 2) Reciprocal
- 3) power (x^2 / x^{x+1})

custom

- 1) Box-Cox
- 2) Yeo-Johnson

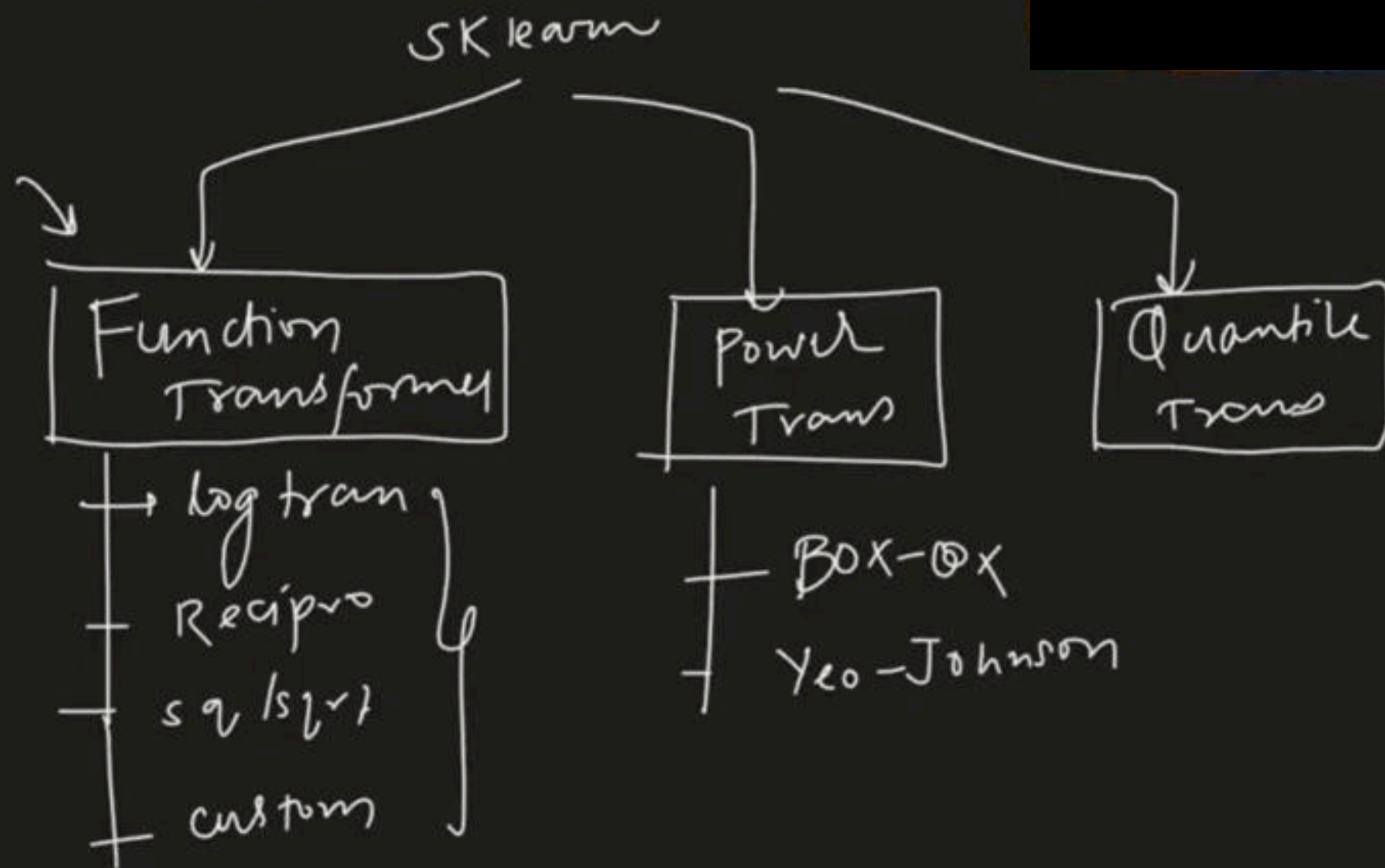
Statistic

ML

Linear Reg
Logistic

Function Transformer

Friday, April 16, 2021 5:22 PM



How to find if data is normal?

Friday, April 16, 2021

6:30 PM

`sns.distplot`



`pd.skew()` = 0

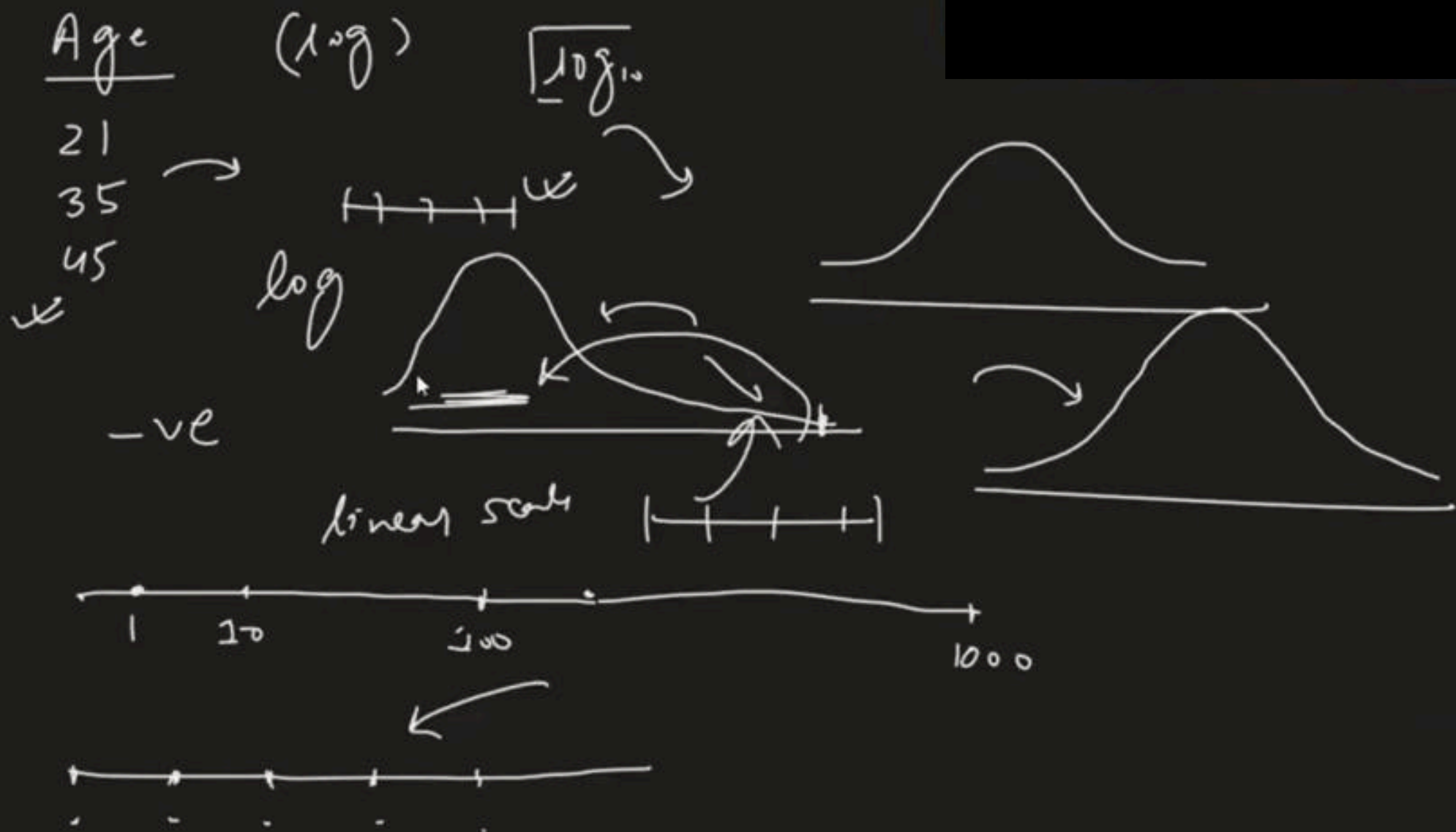
QQ plot

statistics

Log Transform

Friday, April 16, 2021

5:22 PM



Other Transforms

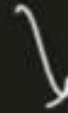
Friday, April 16, 2021

6:23 PM

Reciprocal $1/x$



sq (x^2)



left skewed



sq $x + \sqrt{x}$

sq x^2



Function
Transform

Example

Friday, April 16, 2021 5:22 PM

Titanic



age | fare | survived

np.log

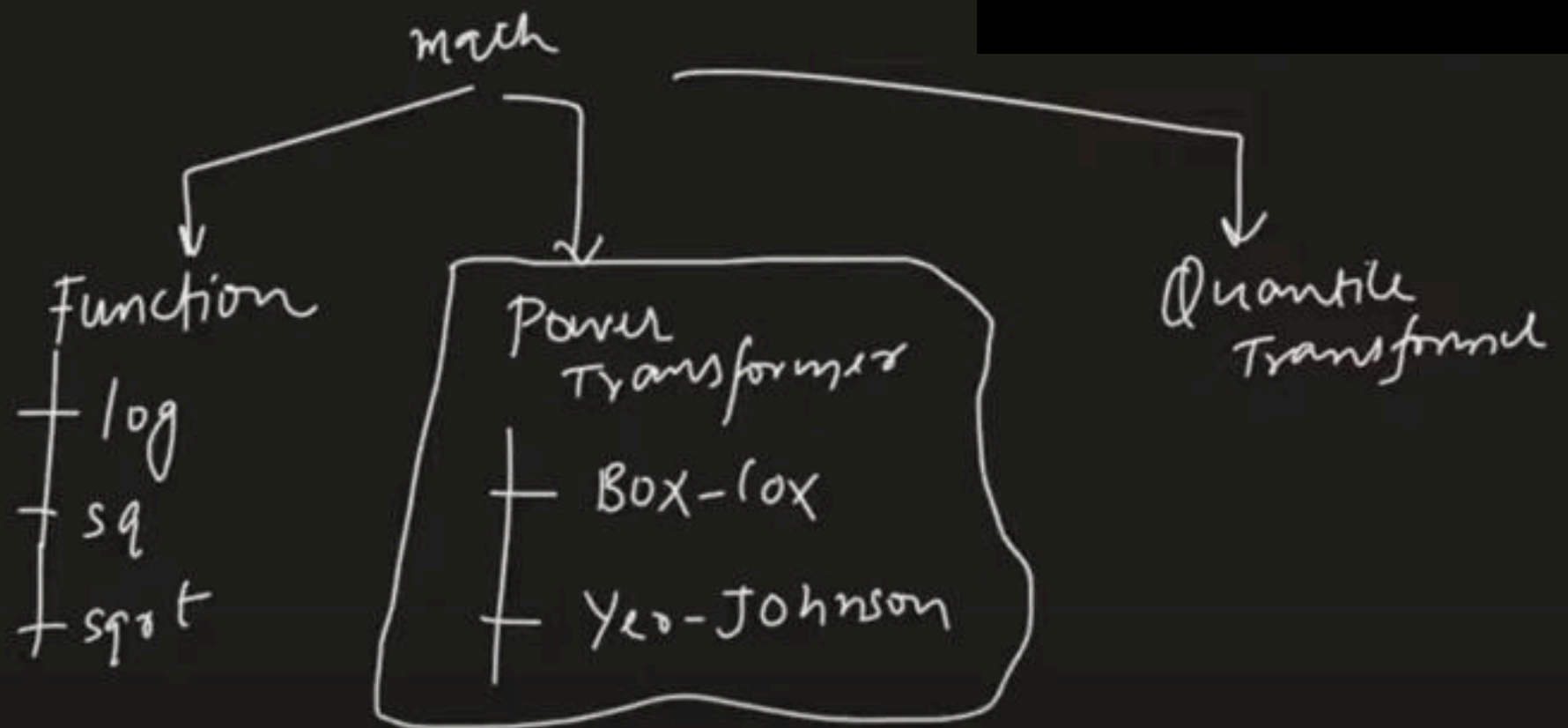
y_0

np.log1p

$(x+1)^*$

Power Transformer

Saturday, April 17, 2021 4:55 PM



Box Cox Transform

Saturday, April 17, 2021 4:56 PM

$$\underline{x_i^{(\lambda)}} = \begin{cases} \frac{x_i^\lambda - 1}{\lambda} & \text{if } \underline{\lambda} \neq \underline{0}, \\ \ln(x_i) & \text{if } \lambda = 0, \end{cases}$$

↳ $\lambda > 0$ (ve)

The exponent here is a variable called lambda (λ) that varies over the range of -5 to 5, and in the process of searching, we examine all values of λ . Finally, we choose the optimal value (resulting in the best approximation to a normal distribution) for your variable.

given dist.
↳ Normal dist

x^2, x^3, x^{15}
1.5, 1.1.

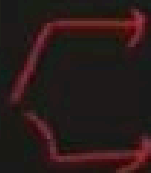
1) Max. likelihood
2) Bayesian

Yeo - Johnson Transform

Saturday, April 17, 2021 4:56 PM


$$x_i^{(\lambda)} = \begin{cases} [(x_i + 1)^\lambda - 1]/\lambda & \text{if } \lambda \neq 0, x_i \geq 0, \\ \ln(x_i) + 1 & \text{if } \lambda = 0, x_i \geq 0 \\ -[(-x_i + 1)^{2-\lambda} - 1]/(2 - \lambda) & \text{if } \lambda \neq 2, x_i < 0, \\ -\ln(-x_i + 1) & \text{if } \lambda = 2, x_i < 0 \end{cases}$$

This transformation is somewhat of an adjustment to the Box-Cox transformation, by which we can apply it to negative numbers.

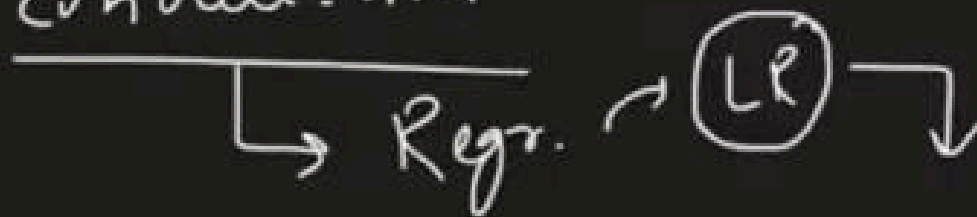
Power Transformation 

Example

Saturday, April 17, 2021

4:56 PM

concrete - unit



cement

$$(540)^{0.177} \rightarrow$$

$$(30)^{(0.02)}$$

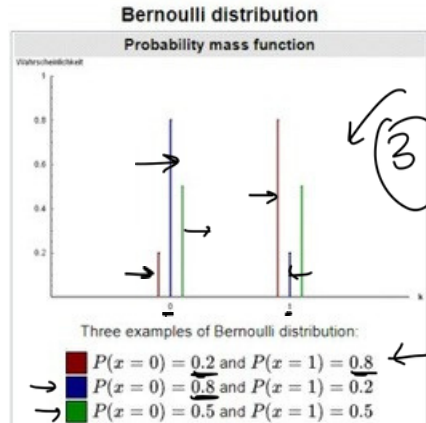
Bernoulli Distribution

27 March 2023 16:06

Bernoulli distribution is a probability distribution that models a binary outcome, where the outcome can be either success (represented by the value 1) or failure (represented by the value 0). The Bernoulli distribution is named after the Swiss mathematician Jacob Bernoulli, who first introduced it in the late 1600s.

The Bernoulli distribution is characterized by a single parameter, which is the probability of success, denoted by p . The probability mass function (PMF) of the Bernoulli distribution is:

$$P(X=x) = p^x (1-p)^{1-x}$$



machine learning

The Bernoulli distribution is commonly used in machine learning for modelling binary outcomes, such as whether a customer will make a purchase or not, whether an email is spam or not, or whether a patient will have a certain disease or not.

Binomial distribution

$p \rightarrow$ prob of success $\rightarrow \hat{p}$
 $1-p \rightarrow$ prob of failure
 Experiment

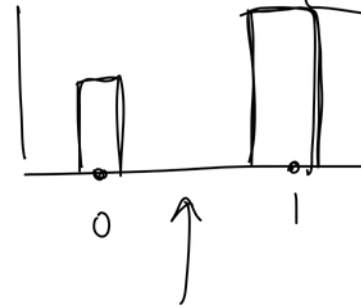
coin toss $\rightarrow$
 binary outcome

$$P(X=1) = \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^0 = \frac{1}{2}$$

$$P(X=0) = \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{1-0} = 1 \times \frac{1}{2} \rightarrow \frac{1}{2}$$

rolling a dice getting a 5

$$\left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^{\sim 5}$$



Binomial Distribution

27 March 2023 16:36

(p) (n, p) $\eta = 1 \rightarrow$ binomial $\downarrow$ bernoulli

Bernoulli trial $\frac{n \text{ times}}{(5)} \rightarrow$ Binomial

Binomial distribution is a probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials with two possible outcomes (often called "success" and "failure"), where the probability of success is constant for each trial. The binomial distribution is characterized by two parameters: the number of trials n and the probability of success p .

Feedback $\rightarrow$ 10 students

0.5

The Probability of anyone watching this lecture in the future and then liking it is 0.5. What is the probability that:

1. No-one out of 3 people will like it

$\frac{1}{8}$

1. 1 out of 3 people will like it

$\frac{3}{8}$

1. 2 out of 3 people will like it

$\frac{3}{8}$

1. 3 out of 3 people will like it

$\frac{1}{8}$

Binomial

$$P(X=x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

$n \rightarrow$ # of trials

$p \rightarrow$ prob of success

$x \rightarrow$ desired result.

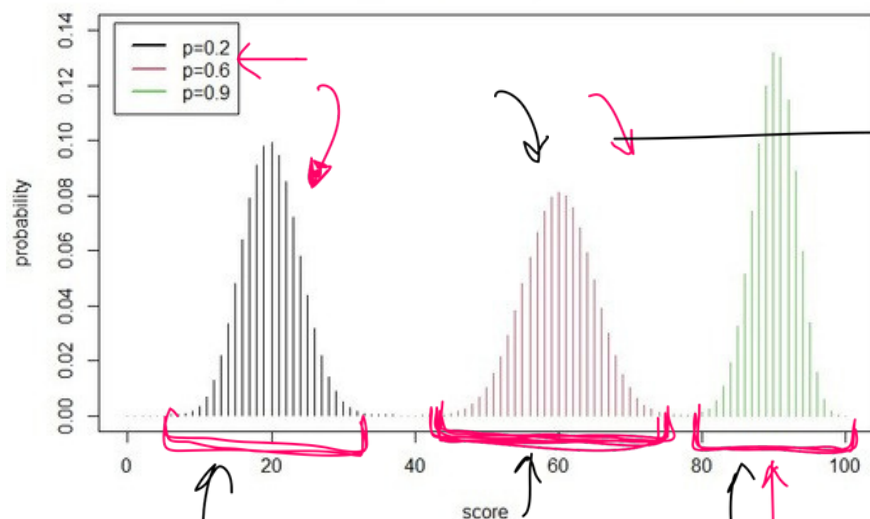
PDF Formula:

$$\frac{3!}{2!1!} \times \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = 3 \times \frac{1}{8} = \frac{3}{8}$$

Graph of PDF:

H T H T H T _ _ 10
H _ _ _ _ _

Binomial distribution with different probabilities of success



$p(s)$ increase

Criteria:

1. The process consists of n trials
2. Only 2 exclusive outcomes are possible, a success and a failure.
3. $P(\text{success}) = p$ and $P(\text{failure}) = 1-p$ and it is fixed from trial to trial
4. The trials are independent.

1. **Binary classification problems:** In binary classification problems, we often model the probability of an event happening as a binomial distribution. For example, in a spam detection system, we may model the probability of an email being spam or not spam using a binomial distribution.

2. **Hypothesis testing:** In statistical hypothesis testing, we use the binomial distribution to calculate the probability of observing a certain number of successes in a given number of trials, assuming a null hypothesis is true. This can be used to make decisions about whether a certain hypothesis is supported by the data or not.

3. **Logistic regression:** Logistic regression is a popular machine learning algorithm used for classification problems. It models the probability of an event happening as a logistic function of the input variables. Since the logistic function can be viewed as a transformation of a linear combination of inputs, the output of logistic regression can be thought of as a binomial distribution.

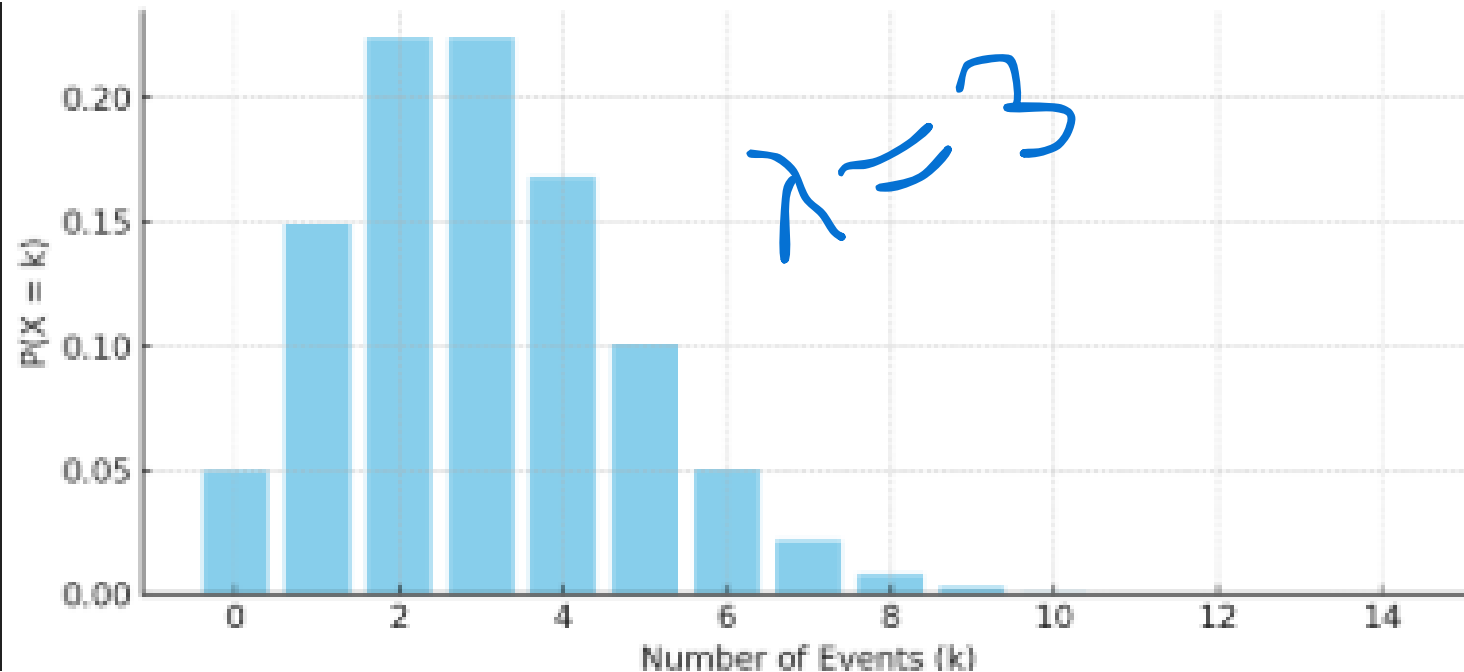
4. **A/B testing:** A/B testing is a common technique used to compare two different versions of a product, web page, or marketing campaign. In A/B testing, we randomly assign individuals to one of two groups and compare the outcomes of interest between the groups. Since the outcomes are often binary (e.g., click-through rate or conversion rate), the binomial distribution can be used to model the distribution of outcomes and test for differences between the groups.

Poisson Distribution

The Poisson distribution is a probability distribution that describes the number of times an event occurs in a fixed interval of time, space, or area, given that the events occur independently and at a constant average rate.

Real-Life Examples:

- Number of emails you receive per hour
- Number of calls a call center gets in a minute
- Number of accidents at an intersection in a day
- Number of decay events per second from a radioactive source



♦ Formula:

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

Where:

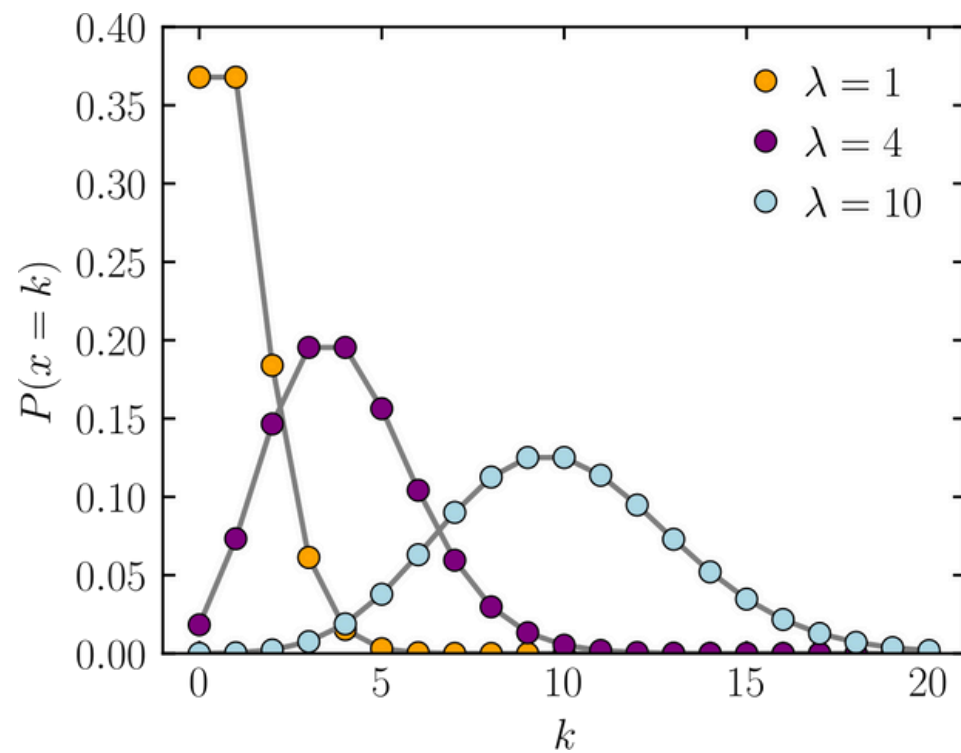
- $P(X = k)$ = probability of observing k events in the interval
- λ = average rate (mean number of occurrences)
- k = actual number of occurrences (0, 1, 2, ...)
- $e \approx 2.718$ (Euler's number)

Mean and Variance:

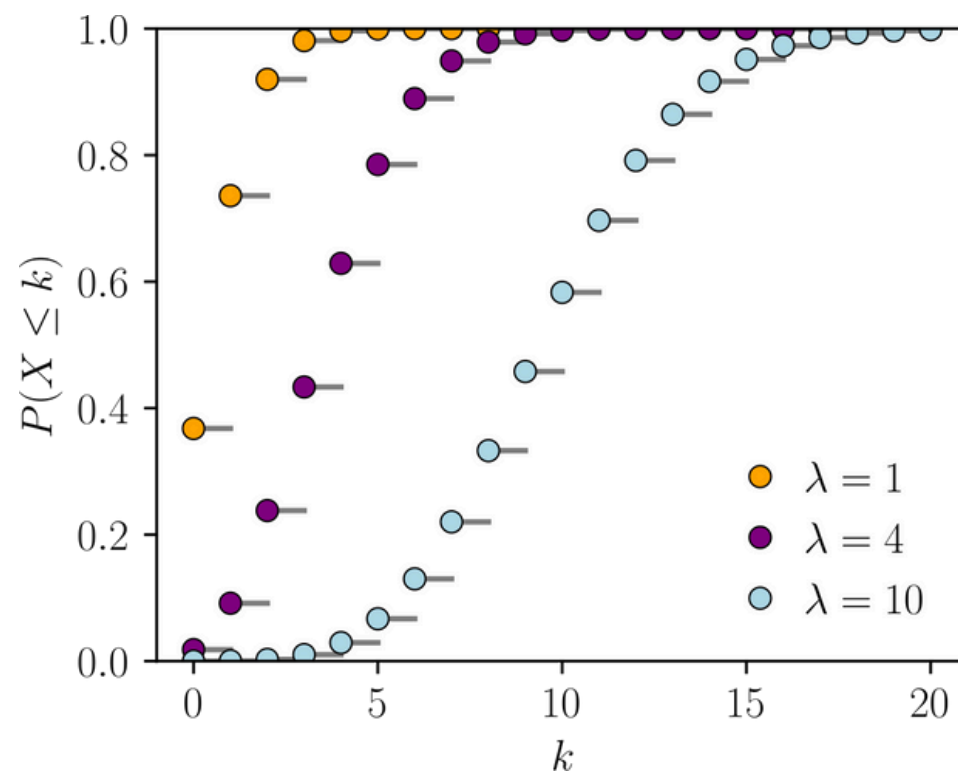
Both the mean and variance of a Poisson distribution are equal to λ .

$$\lambda = \mu = \sigma^2$$

PMF



CDF



Key Assumptions:

1. Events are independent.
2. Events occur at a constant rate.
3. Two events can't happen at exactly the same instant.
4. The average number of events (λ , "lambda") in the interval is known and fixed.